Let Trunk 2 be False. ∴ The treasure is not in Trunk 1.
Since Trunk 1 is True, the treasure is in Trunk 3
Since Trunk 3 is True, it is empty. But this is a contradiction.

∴ Trunk 1 and Trunk 3 cannot both be True at the same time.

Let Trunk 3 be False. ∴ The treasure is in Trunk 3
Since Trunk 2 is True, the treasure is in Trunk 1. This is a contradiction

∴ Trunk 3 must be True

∴ The Queen can say this, and the treasure is in Trunk 1

d) Let all three inscriptions be true
Trunk 1 and 2 contradict each other by respectively stating that the treasures are in Trunk 3 and Trunk 1.

∴ All three inscriptions are True is a False statement, and the Queen cannot say this

18. Suppose that in Example 7 there are treasures in two of the three trunks. The inscriptions on Trunks 1, 2 and 3 are "This trunk is empty," "There is a treasure in Trunk 1," and "There is a treasure in Trunk 2." For each of these statements, determine whether the Queen who never lies could state this, and if so, which two trunks the treasures are in.

a) "All the inscriptions are false."
b) "Exactly one of the inscriptions is true."
c) "Exactly two of the inscriptions are true."
d) "All three inscriptions are true."

Soln: (a) Let all the inscriptions be False. ∴ By Trunk 1's statement, being False the treasure is in Trunk 1. Trunk 2 states that the treasure is in Trunk 1 which is True. This is a contradiction to the fact that all the inscriptions are False. ∴ The Queen cannot say this because all the inscriptions can't be False.